

Karina Maharani

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Portfolio : karinamaharani.github.io

PROFESSIONAL SUMMARY

Final-year Information Systems student ranked in the top 1% of cohort, with interests in AI-enabled healthcare, distributed systems, and computational simulation and modelling. Has a strong ability to understand technical systems in depth while considering their legal, ethical, social, and societal implications. Rigorous, adaptable, and fast-learning, with a commitment to developing innovative technology that is grounded in real-world needs.

Interests: Computational Modelling and Simulation, Digital Health Systems, Federated Learning, Multimodal Learning, Data and Artificial Intelligence Ethics

EDUCATION

Bachelor of Computer Science, University of Indonesia 2023 - Present

- Information Systems Major
- Cumulative GPA: 3.96 / 4.00 (Top 2 of 200 students)
- Expected early graduation (3.5 years)

Incoming Participant for Short Course, Tsinghua University 2026

- Selected as sole Indonesian representative for “The World Nuclear Industry Today” Short Course organized by Tsinghua University in collaboration with the World Nuclear University held on 4 - 10 July 2026
 - Program costs were fully funded by the Asian University Alliance
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RESEARCH EXPERIENCE

Student Researcher Computer Systems Lab University of Indonesia 2026

- Research Title : Clustered Federated Learning with Low Rank Adaptation for Cross-Silo Healthcare
 - Supervisor : Muhammad Hilman, PhD. and Hendrie Gunawan
 - Designed and developed a clustered federated learning model integrated with Low-Rank Adaptation (LoRA) to simulate tiered healthcare systems using Flower framework
 - Developed dataset pipelines integrating symptoms, diagnosis, and geographic data for healthcare pathway recommendation
 - Evaluating model performance to focus on real-world deployment challenges, including non-IID data, system scalability, and model usability in practical healthcare settings
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PUBLICATIONS & WORKING PAPERS

Clustered Federated Learning with Low-Rank Adaptation for Cross-Silo Healthcare (Ongoing, 2026)

TEACHING EXPERIENCE

Teaching Assistant for Discrete Mathematics II, University of Indonesia 2025
Teaching Assistant for Calculus I, University of Indonesia 2024

SKILLS AND COMPETENCES

Python, Java, PyTorch, Scikit-learn, Hugging Face, Flower Framework (Federated Learning), Pandas, NumPy, SPSS, Git, Docker, Figma, Canva

AWARDS & HONOURS

Indonesia Winner for Microsoft ASEAN AI for Accessibility Hackathon 2026
Amartha STEAM Fellowship (Merit-based scholarship for outstanding female undergraduates) 2025
Best Delegate Singapore Model United Nations UNODA Council 2025

INDUSTRY EXPERIENCE

Product Manager Intern, Social Circles Co. 2026

- Improved product flows, UI/UX design, and feature sets, increasing user engagement
- Conducted QA across multiple device architectures to ensure app stability and compatibility
- Ensured project deadlines are met through leading daily stand-up meetings with engineer team
- Integrated OpenAI API to build personalized LLM games, translating AI into product features

Product Innovation and Development Intern, PT Paragon Technology & Innovation

2024

- Researched global and regional beauty and skincare products and markets
 - Conducted market research and competitor analysis for opportunities and consumer preferences
 - Developed decks emerging trends, unique packaging, and innovative formulas for cosmetics
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VOLUNTEERING & LEADERSHIP EXPERIENCE

Head of Musical Development, Paragita Varsity Choir

2025 - Present

- Drafted monthly choir scheduling for rehearsals, concerts, annual programs
- Managed legal and administrative matters related to music performances, including copyright clearance, licensing, and contract documentation
- Designed and led the audition process for competition team selection, including criteria development, scheduling, and communication
- Conducted the first round of audition process for new choir members, including scheduling, communication, and criteria development for 200+ applicants

Emo-Check Social Project Organizer and Financial Controller, Amarthia Foundation

2025 - Present

- Conducted field research and project sensing with a local youth foundation to identify emotional support needs among children from vulnerable communities
- Designed and coordinated a structured social program delivering weekly emotional management activities
- Oversaw finances by managing income, expenses, and budget allocation to ensure effective use of financial aid

English Facilitator for Asian Medical Students' Association

2025

- Guided active medical students in focus group discussion regarding Emerging Technology for Healthcare such as genetic editing and legal compliance for technologies
- Gave feedback to participants post-debate for technology-centric topic in healthcare and debate conduct

Participant of WHO TDR Implementation Research Program

2025

- Participated in discussions regarding ongoing Implementation Research alongside post-doctoral students
 - Developed Implementation Research abstract for E-Health Challenges in Indonesia
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PAST PROJECTS

VoiceBox - Real-time Multimodal Speech Restoration App (available in portfolio site)

Team Leader, Backend Developer, Product Manager | Indonesia Winner Microsoft AI4A Hackathon

- Designed a multimodal speech restoration pipeline combining visual speech recognition, audio speech recognition, LLM sentence correction, and TTS to support both partial and complete voice loss
- Developed user requirements from user interview with potential end users in Dr Cipto Mangunkusumo Hospital Larynx and pharynx ENT polyclinic
- Built product features including dual capture modes, quick phrases, personalized vocabulary history, smart word suggestions, and post-prediction editing before speech output.

HappyHealthy - Curative-centric Mobile Application for Psychiatric Medication (available in portfolio site)

- Designed a mobile health concept to support psychiatric medication adherence, centered on drug-drug interaction checking, medication reminders, side-effect logging, and caregiver monitoring with risk alerts
 - Specified functional, non-functional requirements, including DrugBank API integration for interaction checking, Database EERD schema, Use-Case Diagram, Use-Case Description, Sequence Diagram
 - Conducted early user research, user acceptance utilizing usability testing methodology, and SUS Questionnaire with real end-user
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LANGUAGE CERTIFICATION

English - IELTS Academic, Overall Band 8.5 (CEFR C2)

Indonesian - Ujian Kemahiran Bahasa Indonesia, Tingkat Istimewa (CEFR C2); native proficiency